FINM3013 Introduction to Financial Derivatives

Chapter 06

Solution to Class exercises

1. C. The cash price is the quoted price plus accrued interest. There are 30 actual days between April 1 and May 1 and 183 actual days between April 1 and October 1. In this case the quoted price is 105 and the accrued interest is $0.06\times100\times30/183=0.98$. The answer is therefore 105.98.

2. C. The cost of delivering a bond is the quoted bond price minus the most recent settlement price times the conversion factor. This is 2.36, 2.68, 1.625, and 3.275 for bonds in A, B, C, and D, respectively. The bond in C is therefore cheapest to deliver.

3. B. The trader gains $25 for each basis point. The gain is therefore 25×6 or $150.

4. D. The duration of the first bond is 5 years and the duration of the second bond is 10 years. The duration of the portfolio is a weighted average with weights corresponding to the amounts invested in the bonds. It is $0.2\times5+0.8\times10=9$ years.

5. B. The futures rate must be reduced by what is known as a convexity adjustment to get the forward rate.